

Study Notes

This document appears to be a resume.

Study Notes: Sushmita D - Aspiring Software Development Engineer

I. Career Objective

* **Aspiring Software Development Engineer** with strong foundations in:

- * Java, Python, SQL
- * Data Structures & Algorithms
- * Full Stack Development
- * **Experienced in designing**:
- * REST APIs
- * Backend services
- * Scalable software solutions
- * **Technologies used**:
- * Spring Boot, React.js, FastAPI, PostgreSQL, Redis, Docker
- * **Passionate about**:
- * Building reliable systems
- * Solving complex engineering problems

**II. Technical Skills

* **Languages**:

- * Python
- * Java
- * SQL
- * **CS Fundamentals**:
- * **Data Structures & Algorithms**
- * **Object-Oriented Programming**
- * Database Management Systems
- * Operating Systems
- * Computer Networks
- * Software Engineering Principles
- * **Frameworks & Technologies**:
- * **Spring Boot**
- * **React.js**
- * **FastAPI**
- * Flask
- * **REST APIs**
- * **Databases**:
- * MySQL
- * **PostgreSQL**
- * **Redis**
- * MongoDB
- * **Cloud & DevOps**:
- * AWS (EC2, S3)
- * Git
- * **Docker**
- * CI/CD Fundamentals
- * Postman
- * **AI Technologies**:
- * **GenAI**
- * **Agentic AI**
- * **LLMs**
- * **RAG**

III. Internship Experience

- * **KodNest – Full Stack Development Intern** (Feb 2026 – May 2026)
- * Developed **full-stack applications** using Java, Spring Boot, React.js, SQL, and REST APIs.
- * Gained hands-on experience in backend development, debugging, integration testing, and software engineering practices.
- * **MindMatrix.io – Generative AI Intern** (Feb 2026 – May 2026)
- * Worked on **Generative AI** and **Agentic AI applications**.
- * Involved **LLM-powered workflows**, **RAG-based systems**, API integrations, and intelligent automation.
- * Followed software engineering and Git-based development practices.

**IV. Projects

- * **WellKart AI Try-On Platform**
- * **Technologies**: Spring Boot, FastAPI, PostgreSQL, Redis, Docker
- * **Description**:
 - * Designed and developed **backend services** and **REST APIs** following **microservice architecture principles**.
 - * Developed service integrations and containerized workflows to support scalable, maintainable, and high-performance applications.
- * **Hand Gesture Presentation Controller**
- * **Technologies**: Python, OpenCV, MediaPipe, PyAutoGUI
- * **Description**:
 - * Implemented **MediaPipe hand landmark detection** and OpenCV-based frame processing for low-latency real-time gesture recognition (95%+ identification accuracy).
 - * Integrated **PyAutoGUI** to map finger gestures into slide navigation actions, enabling low-latency touch-free presentation control and improving interaction efficiency by 40%.
- * **Black Pepper Adulteration Detection System**
- * **Technologies**: Python, Flask, YOLOv8, MobileNetV2
- * **Description**:
 - * Built a **web-based adulteration detection application** using Python, Flask, and deep learning models for automated quality assessment.
 - * Deployed the application on Hugging Face, enabling real-time image-based prediction and user interaction.

**V. Education

- * **Bachelor of Engineering (CSE)**: 2022 – 2026
- * Srinivas Institute of Technology, Mangalore
- * CGPA: 8.0

**VI. Certifications

- * **IT Specialist – Artificial Intelligence** (Certiport)
- * **Machine Learning Fundamentals Micro-Credential** (Alteryx)
- * **Python Essentials 2** (Cisco)
- * **Big Data Computing** (NPTEL)
- * **IT Specialist - Cloud Computing** (Certiport)
- * **No SQL - MongoDB** (IBM)

**VII. Achievements

- * **Published Research Paper**: "Hand Gesture-Based Presentation Control System," JETIR Journal